MIL-C-16065B(SHIPS)
15 September 1966
SUPERSEDING
MIL-C-16065A(SHIPS)
26 March 1965
(See 6.3)

MILITARY SPECIFICATION

COMPUTER, LONGITUDINAL BENDING STRESS AND SHIP'S DRAFT

1. SCOPE

1.1 This specification covers a longitudinal stress and ship's draft computer, a calculating device which automatically predetermines the nominal bending stress and fore and aft draft which a ship will have under any longitudinal distribution of loading.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-983 - Interior Communication Equipment; Naval Shipboard, Basic Design Requirements for.
MIL-M-15071 - Manuals, Equipment and Systems.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.2).

3.2 General features. - The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern):

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Springs (material)
Other metals
Plastics
Ceramics
Glass
Lubricants and lubrication
Painting (see 3.5)

FSC 6605

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Protection against corrosion
Bolts, machine screws, studs and nuts
Parts - mechanical
Locking devices
Washers
Bearings, ball
Enclosures - general
Enclosure - accessibility
Enclosure - degree of (dripproof)
Enclosure - mounting
Stiffening grooves
Threaded devices
Rounded corners and edges
Drilled and tapped holes
Structural welding
Temperature and humidity
Inclination
Drawings - preliminary
Drawings - working
Bill of materials
Designation and marking
(Serial numbers are required)
Item names and nomenclature
Workmanship

3.3 Design features. -

3.3.1 The computer shall be as compact and lightweight as practicable and shall be suitable for bulkhead mounting (dripproof enclosure) aboard ship. Weight shall not exceed 25 pounds and size 29 inches by 18 inches by 6 inches (approximate).

3.3.2 Reliability. - The reliability of the computer must meet the requirements of Naval service. It is imperative that reliability of operation be considered of prime importance in the design and manufacture of the computer. The contractor shall employ all methods possible in the process of manufacture which will assure quality and maximum reliability consistent with the state of the art. The equipment shall represent the simplest design consistent with the functional and environmental requirements specified herein.

3.3.3 Electrical power. - No electrical power shall be required to operate the computer. The computer shall be completely independent of internal or external electrical power and the displayed information shall be clearly visible and readily interpreted from a distance of 18 inches.

3.3.4 The computer shall have one input actuator for each longitudinal load station on which the respective load shall be set and displayed so that the input actuators jointly form a status board of the longitudinal load distribution. The input scales shall be readable to 1/1000 of the ship's total dead weight.

3.3.5 The computer shall compute and indicate automatically the following outputs:

(a) Total bending stress for both hogging and sagging conditions.
(b) Ship's draft, forward and aft.

3.3.5.1 When used in accordance with the instructions, the computer shall indicate stress values obtained from Hogging and Sagging Stress Tables developed by the Naval Ship Engineering Center (NAVSEC) for the particular ship to which the computer is applicable within a tolerance of two-tenths ton per inch square in the range from 4 to 9 tons per inch square and from full load to medium or one-half load.

3.3.5.2 The draft scales should be calibrated in accordance with information given on Curves of Form and Inboard Profile Drawings applicable to each specific vessel or class of vessels for which the computer is applicable. The scales shall be graduated in feet and half-foot intervals; the reading shall be accurate to within a tolerance of 1-1/2 percent of full draft at level keel, and of 2-1/2 percent of full draft at 20 foot trim.

3.3.5.3 Information pertinent to the development of scales and data readout requirements shall be provided by the cognizant Supervisor of Shipbuilding or NAVSEC (see 3.3.5.1 and 3.3.5.2).

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3.4 Operating. - An operating test shall be conducted for each computer as specified in 4.4.

3.5 Painting. - The front panel shall be of a natural anodized aluminum color with lines and letters engraved and filled with suitable paint. The case shall be painted with a grey finish. The name or type of vessel shall appear on the front panel of each computer.

3.6 Manuals. - Manuals shall be type I of MIL-M-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - Preproduction inspection shall consist of the examination and tests listed in table I, which shall be performed in general in the order listed.

Table I - Preproduction inspection.

Test	Requirement paragraph	Test reference
General examination	---	MIL-I-983
Operating	3.4	4.4
Temperature and humidity	3.2	MIL-I-983
Inclination	3.2	MIL-I-983
Enclosure	3.2	MIL-I-983

4.3 Quality conformance inspection. - All computers offered for delivery on the contract or order shall be subjected to the examination and test specified in table II. Any equipment failing to meet the requirements of this specification shall not be offered for delivery.

Table II - Quality conformance inspection.

Inspection	Requirement paragraph	Test reference
General examination	---	MIL-I-983
Operating	3.4	4.4

4.4 Operating test. - The operating test shall consist of five bench trials each simulating a hypothetical loading operation. Total bending stress and draft indications for each trial test shall be determined via the computer and resulting data compared with those extracted from Hogging and Sagging Stress Tables and draft Curves of Form and Inboard Profile Drawings furnished by NAVSEC, for identical trial loading conditions. Tolerance deviations in excess of tolerances specified in 3.3.5 must be recorded and subsequently resolved to the satisfaction of the cognizant Government inspector.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Computers. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, manufacturer's name and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation, packaging, packing and marking. - The equipment and accessories, and technical publications shall be preserved and packaged levels A or C; packed level A or B and marked in accordance with MIL-E-17555, as specified.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Quantity of manuals required.
(d) Preservation and packaging, packing and marking if other than as specified in 5.1 (see 5.2).

6.2 Preproduction. - Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project-6605-N107Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE D14000

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Naval Ship Engineering Center

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